

MOTIVATION

Galois $K/\mathbb{Q}$ compute $\text{Gal}(K/\mathbb{Q})$

K any field $G_K = \text{Gal}(K^{\text{sep}}/K)$

Inverse Galois problem

G finite group what is $\mathbb{Q}$ field K
such that $G \cong \text{Gal}(K/\mathbb{Q})$

G profinite group find K

if $|G_K| = 1$ then $K = K^{\text{sep}}$

$|G_K| = 2$ then K is real closed

- -1 is not a sum of squares (real)
- maximal real

examples $\mathbb{R}$, $\mathbb{R} \cap \mathbb{Q}^{\text{sep}}$, $\bigcup_{n \in \mathbb{N}} \mathbb{R}((t^{1/n}))$

in general if $G_K \cong G_L$ for
free is no hope to have $L \cong K$

but we could have $L \cong K$

K and L share the same 1st order properties

not 1st order $\rightarrow$ cardinality

$\rightarrow$ Archimedean property

$\forall x, y \in \mathbb{R} \exists x < y \exists n \in \mathbb{N}$ such

not $y < nx$

(Königsmann) if $G_K \cong G_{\mathbb{Q}_p}$
then $K \cong \mathbb{Q}_p$

example $\mathbb{Q}^{\text{sep}} \cap \mathbb{Q}_p = \bigcup_n \mathbb{Q}_p((t^{1/n}))$

Artin/Schreier / Ax-Kochen

$K \cong \mathbb{Q}_p$ iff K admits a valuation v st.
 $\text{char}(K) = 0$
 $K_v \cong \mathbb{F}_p$
 $v(p)$ is min positive (e.g. $\mathbb{Q}$)

K is maximal such

$\Leftrightarrow v$ is henselian

$\bullet v(K/(v(p)))$ is divisible

(equivalently $v(K) \cong \mathbb{Z}$)

e.g. $\mathbb{Q} \times \mathbb{Z} \cong \mathbb{Z}$

actually all of these are

Smith divisible $\times \mathbb{Z}$

Goal of today: prove that $G_{\mathbb{Q}_p}$ is "small"
i.e. $\mathbb{Q}_p$ has finitely many alg. extensions
of deg $n \forall n$

($\rightarrow$ Corollary 3.17)

DEFINITION 3.1

F a field. We denote by

$$\mu_F = \{x \in F \mid x^n = 1 \text{ for some } n\}$$

$$\mu_n = \{n\text{-th root of unity in } \overline{F}^{\text{alg}}\}$$

ζ_n = primitive n -th root of unity

$$\mathbb{Q}_p^\times = \bigcup_{n \geq 1} \mathbb{Q}_p^{\times n}$$

we always assume compatibility $\mathbb{Q}_p^\times = \bigcup_{n \geq 1} \mathbb{Q}_p^{\times n}$

LEMMA 3.2

$L = \mathbb{Q}_p(\zeta_n)$ • if $(n, p) = 1$ then $L/\mathbb{Q}_p$ is
unramified and

$$\text{Gal}(L/\mathbb{Q}_p) \cong \text{Gal}(\mathbb{F}_p(\zeta_n)/\mathbb{F}_p)$$

$$\cong \mathbb{Z}_k$$

where $k = \text{ord}_n(p)$

(as $p \in (\mathbb{Z}/n\mathbb{Z})^\times$)

(cyclic group
of order k)

• if $n = p^m$ then $L/\mathbb{Q}_p$ is totally ramified
of degree $(p-1)p^{m-1}$

$$\text{and } \text{Gal}(L/\mathbb{Q}_p) \cong (\mathbb{Z}/p^n\mathbb{Z})^\times$$

COROLLARY

if $n \mid p-1 \Rightarrow \zeta_n \in \mathbb{Q}_p$ for p odd

$$\mu_{\mathbb{Q}_p} = \begin{cases} \mu_{p-1} & \text{if } p \text{ odd} \\ \mu_2 & \text{if } p = 2 \end{cases}$$

LEMMA 3.4

let (K, v) henselian valued field
and let q be a prime

Assume (i) $\text{char}(K) \neq q$

(ii) v is not q -divisible

$$(iii) \text{Gal}(K_v(q)/K_v) \cong \mathbb{Z}_q$$

$K_v(q)$ is h.c. q -closure

$\Rightarrow \mathbb{Q}_p$ is fine
and also
all finite
extensions of $\mathbb{Q}_p$

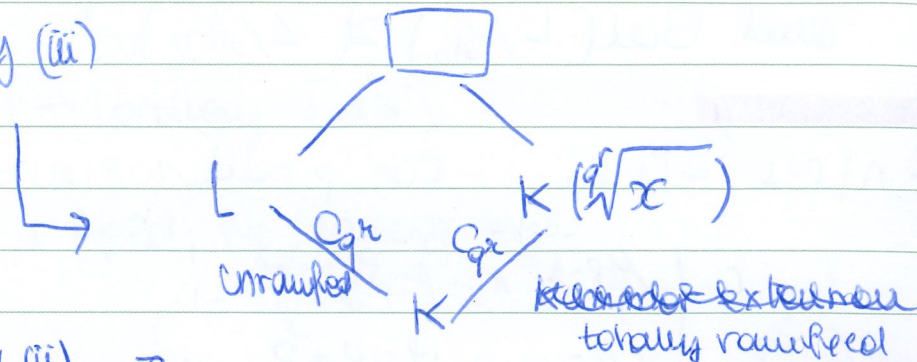
Then $\zeta_{q^k} \in K$ iff K admits a
 $C_{q^k} \times C_{q^k}$ extension

proof

$\Rightarrow$ Assume $\rho_{qr} \in K$

by (ii)

this is our extension



by (ii) with
for $x \in K$, $\rho(x) \in K$ not q -divisible

by Kummer theory

$\Leftarrow$ Let L/K be a $C_{qr} \times C_{qr}$ extension

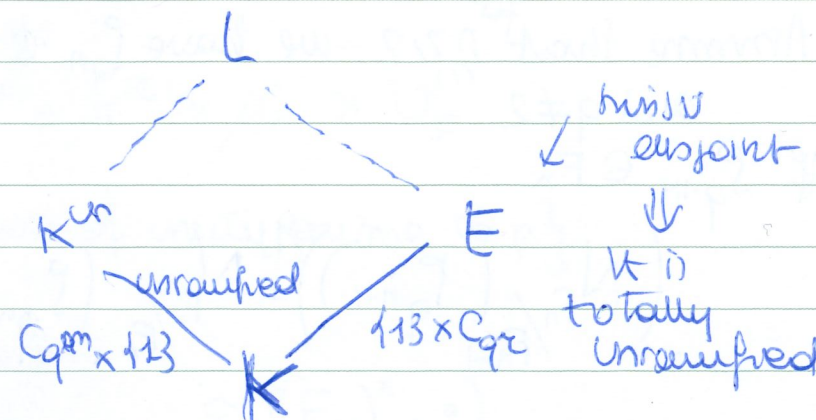
Let $K^{ur} =$ maximal unramified ex in L

$\Rightarrow$ it is a q -extension

all extensions of degree a power of q
are cyclic by (iii)

because the Galois group is a quotient
of $C_{qr} \times C_{qr}$

$\Rightarrow$ we are inside one component of $Q_r \times Q_r$



this is disjoint

it is totally unramified

$$\Rightarrow \text{Gal}(E/K) \cong I(E/K) \cong C_{qr} \\ \cong \text{Hom}(\bigvee_{v \in K} \mu_{E_v})$$

we have q -torsion $\Rightarrow \rho_{qr} \in \mu_{E_v}$
(since $E_v = K_v$)

LEMMA 3.6

Let F be a field with $\text{char } F \neq q$

Then $F(\rho_{q^\infty})/F$ is either infinite

in which case $\text{Gal}(F(\rho_{q^\infty})/F) \cong \mathbb{Z}_q \times C$

where C is cyclic of order dividing $q-1$

or it is finite in which case $F(\rho_{q^\infty}) = F(\rho_q)$
for q odd and $F(\rho_{2^\infty}) = F(\rho_4)$ otherwise

proof $F_n = F(\zeta_{q^n})$ for $n \in \mathbb{N}$

Assume that ^{for} $n \geq 2$ we have $\zeta_{q^n} \notin F_{n-1}$

and $q \neq 2$

if $\zeta_{q^{n+1}} \in F_n$

$$(N_{F_n/F_{n-1}}(\zeta_{q^{n+1}}))^q = N_{F_n/F_{n-1}}(\zeta_{q^n}) = \zeta_{q^{n-1}}$$

with $\zeta_{q^n} \notin F_{n-1}$

$\Rightarrow F_n$ is an infinite degree tower

NEW GOAL

$(F^x)/(F^x)^q$ we want to understand its dimension for $F/\mathbb{Q}_p$ finite

$$F^x \supseteq U_F^{(0)} \supseteq U_F^{(1)} \supseteq \dots \supseteq U_F^{(k)} \supseteq \dots$$

DEF 3.7

$$U_F^{(0)} = \mathcal{O}_F^x$$

$$U_F^{(k)} = \{x \in \mathcal{O}_F^x \mid x \equiv 1 \pmod{\pi^k}\}$$

for $k \geq 1$

~~exp~~

FACT 3.8

$$F^x = \pi^{\mathbb{Z}} \times \mu_F^1 \times U_F^{(1)}$$

$$\mu_F^1 = \{\text{roots of unity prime to } p\}$$

FACT 3.9

$$U_F^{(0)}/U_F^{(1)} \cong ((F_v)^x, \cdot)$$

$$U_F^{(k)}/U_F^{(k+1)} \cong (F_v, +)$$

proof

$$\begin{aligned} x U_F^{(1)} &\mapsto x + M_F \\ (1 + \pi^k a) U_F^{(k+1)} &\mapsto a + M_F \end{aligned}$$